

Resubmission note

The Lock-Step Tax: Per-Lane Control Flow as the Missing Primitive in Tile GPU DSLs

Previous submission. This work was submitted once before, to ACM TACO, on 5 July 2026. It was returned by the Editor-in-Chief on 7 July 2026 as an editorial decision, without external review and without reviewer reports. The stated grounds were that the manuscript did not meet the journal’s novelty and quality bar, and that the results were “too premature for publication,” with more work needed to substantiate the conclusions. It has not been submitted to ASPLOS before, nor to any other venue, and it is under review nowhere else.

How the work changed since. Because no reviewer reports were provided, we treated the one substantive signal we had, prematurity, as the thing to fix. The weakest claim at the time was the cure: it existed as a single compiler build on a single GPU, which is exactly the shape of a result that cannot yet carry a conclusion. Four changes followed.

1. *The cure is now cross-architectural and rebuildable.* The TritonGPU→LLVM per-lane-retirement pass is rebuilt from a pinned one-command recipe, and the speedup reproduces on a second architecture: 1.6 to 3.8× on an A100, against 2.3 to 6.7× on the RTX 4070, from the same artifact. The cure is no longer one machine and one build.
2. *Numbers that could not be reproduced from the pinned recipe were replaced.* During the rebuild we found that some figures came from an earlier local build that the pinned recipe did not reproduce exactly. Those numbers were re-measured and replaced rather than carried over.
3. *The predictive claim was refit and tested out of sample.* The straggler law was refit ($R^2 = 0.997$) and validated on held-out configurations, with 1.5% and 2.1% out-of-sample error, so it is now a prediction rather than a fit.
4. *Negative results were added, deliberately.* The paper now carries a cure-rejection witness (a kernel where the pass correctly declines to fire) and a generality law over eight oracle-gated workloads, including a pointer-chase negative control at 1.00× and a case where the sign inverts and the tile wins (0.64×). These bound the claim instead of extending it.
5. *One statistic was replaced with a more robust one.* The claim that the lock-step cost is set by the absolute per-warp straggler rather than by the divergence *ratio* previously rested on a correlation computed over the six designed distributions, where it is 0.06. Adding the four held-out distributions raises it to 0.55, so the statistic was sample-fragile. The claim now rests on the comparison of fits, which is stable on both samples: $R^2 = 0.997$ and 0.998 for the straggler against 0.00 and 0.29 for the divergence ratio. The conclusion is unchanged; the evidence for it is no longer fragile.

Presentation. The manuscript was condensed from the 16-page `acmsmall` journal format to the ASPLOS format and page limit, and then restructured: the paper is retitled after the mechanism it identifies rather than after the workload it was first measured on, the contribution list is consolidated to four items, the abstract is roughly half its former length, and the argument now reaches its central claim, its figure and its contributions within the first two pages. The compiler section was broken into subsections, the figures were rebuilt as vector graphics at their final printed size, and a listing of the matched TritonGPU IR was added so the structural impossibility result can be read off the code rather than taken on trust. An acknowledgments section disclosing the use of a generative AI assistant was added, per the ACM authorship policy. The technical content is otherwise the content described above.